

From lexicon to phrase: Behavioral evidence for a dynamic cognitive mechanism underlying language control

Qingwei Xue^{a,b}, Qiping Wang^b, Junjie Wu^c, Luyao Chen^{b,d,*}

^a School of Chinese as a Second Language, Peking University, Beijing, China

^b Max Planck Partner Group, School of International Chinese Language Education, Beijing Normal University, Beijing, China

^c Key Research Base of Humanities and Social Sciences of the Ministry of Education, Academy of Psychology and Behavior, Tianjin Normal University, Tianjin, China

^d Department of Neuropsychology, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany

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ABSTRACT

This study examined language control at both lexical and phrasal levels, and their relationships with general cognitive control. Thirty-three participants completed two picture-naming tasks assessing lexical and phrasal control. We measured mixing costs (i.e., the index of proactive language control) and switch costs (i.e., the index of reactive language control), and examined their relationships with general proactive cognitive control, assessed by the Proactive Behavioral Index (PBI) from the AX-CPT task, and with general reactive cognitive control, assessed by Stroop scores from the Stroop task. Results revealed greater L1 than L2 mixing costs at both linguistic levels. Switch costs were symmetrical at the lexical level but asymmetrical at the phrasal level. PBI predicted mixing costs at both linguistic levels, whereas Stroop scores predicted switch costs only at the phrasal level. These findings suggest that bilinguals rely on proactive control to monitor bilingual lexical competition, whereas both proactive and reactive control are engaged at the phrasal level. This study proposes a hypothesis that control demands vary across linguistic levels, influencing the adaptation of language control mechanisms and their relationships with general cognitive control.

1. Introduction

Bilinguals, proficient in two or more languages, represent one of the most distinctive groups in contemporary society (Grosjean & Li, 2013; Mehmedbegovic, 2018; Sinkeviciute et al., 2024; Titus et al., 2024). Psycholinguistic evidence suggests that competition exists between the languages of bilinguals (e.g., Abutalebi & Green, 2008; Genesee et al., 1978; Guo & Peng, 2006; Ivanova & Hernandez, 2021; Macizo et al., 2012). To generate the target language, bilinguals frequently engage in language selection and generation, developing a mental mechanism known as **language control**. This mechanism prevents interference from the non-target language and facilitates the selection and generation of the target language (Abutalebi & Green, 2008; Declerck & Philipp, 2015a; Green, 1998; Meuter & Allport, 1999).

Previous research on language control has primarily focused on the lexical level, however, the control mechanisms may extend beyond the lexicons to phrases, sentences, and paragraphs (e.g., Chang et al., 2023; Declerck & Philipp, 2015b; Sánchez et al., 2022; Tarlowski et al., 2013; Titus & Peeters, 2024; Zhang & Wang, 2012). Moreover, language

control at other linguistic levels, compared to the lexical level, involves not only the activation of individual lexical items but also their combination (Chen et al., 2023; Friederici, 2011, 2018; Wu, Cheng, et al., 2024). This process might require greater allocation of cognitive resources and adjustments in language control mechanisms for bilinguals (Titus & Peeters, 2024; Zhang & Wang, 2012). Hence, conclusions drawn from lexical-level research are limited. It is crucial to explore language control at different linguistic levels, such as the lexical and phrasal levels, and to compare these mechanisms across different linguistic levels.

Moreover, an important and widely discussed topic in the field of language control is the relationship between language control and general cognitive control. This exploration addresses the critical question of whether language control is domain-specific or domain-general (Branzi et al., 2016; Cui et al., 2024; De Bruin et al., 2014; Declerck et al., 2017; Jiao et al., 2021, 2022; Kang et al., 2017; Wu et al., 2022). This topic is closely related to another pair of control mechanisms: according to the dual control mechanisms framework, individuals employ two distinct control mechanisms, **proactive control** and **reactive**

* Corresponding author at: Max Planck Partner Group, School of International Chinese Language Education, Beijing Normal University, Beijing, 100875, China.
E-mail address: luyaochen@bnu.edu.cn (L. Chen).

control (Appelbaum et al., 2014; Braver, 2012). Proactive control involves anticipatory regulation of the target events before competition arises, whereas reactive control entails immediate regulation in response to competition, relying on temporary inhibition of the non-target events (Braver, 2012; Peeters & Dijkstra, 2018; Wang et al., 2022; Wu, Ji, et al., 2024). Originally developed based on research in general cognitive control, this framework has since been extended to language control studies (e.g., Declerck, Grainger, & Hartsuiker, 2021; Ma et al., 2016; Seo & Prat, 2019; Wang et al., 2022; Wu, Ji, et al., 2024, 2025; Zhang et al., 2021). Specifically, proactive control engages top-down processes, such as the global and sustained inhibition of the non-target language or continuous monitoring of the target language in anticipation of potential conflict (Appelbaum et al., 2014; Braver, 2012; Seo & Prat, 2019; Zhu & Sowman, 2020). In contrast, reactive control relies on bottom-up mechanisms, involving the immediate inhibition of the non-target language in response to cross-language competition, and is more item- or stimulus-specific in nature (Seo & Prat, 2019; Zhu & Sowman, 2020). These mechanisms are interdependent and work cooperatively, with individuals adjusting between them based on varying control demands (Braver et al., 2009; Braver, 2012; Declerck, Grainger, & Hartsuiker, 2021; Declerck, Meade, et al., 2021; Ma et al., 2016; Seo & Prat, 2019).

Previous research has typically focused on their relationship in a single dimension, that is, the reactive control (e.g., Declerck et al., 2017; Festman et al., 2010; Jiao et al., 2019, 2021; Jylkkä et al., 2018; Linck et al., 2012, 2020; Prior & Gollan, 2013), while paying less attention to their dynamic and adaptive nature (Abutalebi et al., 2013; Burgess et al., 2010; Burgess & Braver, 2010; Jiang et al., 2023, 2024; Ma et al., 2016; Seo & Prat, 2019; Wang et al., 2022). Some existing studies have focused on individual or external factors such as L2 proficiency (Festman et al., 2010; Mouthon et al., 2020; Wang et al., 2022), context (Han et al., 2025; Jiang et al., 2023, 2024), language production modes (Wong & Maurer, 2021; Yang et al., 2024), and suggested that these factors might affect the selection of these control mechanisms and the relationships between the two control domains. However, critical internal factors, such as linguistic levels, failed to be the focus of these studies. Since bilinguals may encounter different control demands at the lexical and phrasal levels, investigating how linguistic levels affect the relationship between language control and general cognitive control could yield deeper insights into the dynamic nature of the cognitive mechanism underlying language control.

1.1. Research on language control: A shift in linguistic complexity

Research on language control primarily utilized the cued language-switching paradigm (e.g., Declerck et al., 2013; Koch et al., 2010; Meuter & Allport, 1999; Ward & Awani, 2024; Wu, Ji, et al., 2025). Bilinguals are instructed to name pictures in either their first language (L1) or second language (L2), depending on the colors of the surrounding frames. This setup generates two conditions: switch and non-switch. Studies consistently demonstrate that bilinguals perform worse under switch conditions compared to non-switch conditions, as evidenced by longer reaction times (RT) and reduced accuracy (ACC) (e.g., Costa & Santesteban, 2004; Declerck & Philipp, 2015a; Meuter & Allport, 1999; Wang et al., 2022; Yang et al., 2024). This phenomenon, termed **switch cost**, reflects the cognitive resources required for reactive language control. A smaller switch cost indicates a higher ability in reactive language control among bilinguals.

The Inhibitory Control Model (ICM) (Green, 1998) forms the foundation of language control research. It posits that bilinguals achieve language control through the inhibition of non-target language words while selecting target language words. According to the ICM, bilingual lexicons are tagged according to their respective languages (L1 or L2). Words from both languages share overlapping conceptual representations and are activated simultaneously during language processing. To effectively select and generate words in the target language, bilinguals

establish language task schemas. This mental system enables them to inhibit non-target language words and maintain the activation of target language words, thereby achieving effective control over both languages.

Building on the ICM, two hypotheses have been proposed: (1) The degree of inhibition applied to a language is proportional to proficiency in that language, and (2) The degree of disinhibition is proportional to the extent of prior inhibition applied to that language. Specifically, bilinguals are generally more proficient in L1 than in L2. Thus, controlling L1 requires greater cognitive resources compared to controlling L2. When a previously inhibited language is reactivated, the cognitive resources needed for disinhibition are proportional to the degree of prior inhibition. In other words, the greater the inhibition, the more cognitive resources are required for disinhibition. This is known as the Non-Specific Language Selection Hypothesis (Bobb & Wodniecka, 2013; Costa et al., 2006; Koch et al., 2010; Philipp & Koch, 2009).

Meuter and Allport (1999) used the cued language-switching paradigm, requiring less proficient bilinguals to name numbers in either their L1 or L2 based on the color of the surrounding frame. The results revealed switch costs in both directions: L1 to L2 and L2 to L1. Notably, the switch cost was greater for switches from L2 to L1 than for switches from L1 to L2, supporting the Non-Specific Language Selection Hypothesis. Specifically, switching from L2 to L1 involves two mental operations: inhibiting L1 during the L2 trial and subsequently releasing it during the L1 trial. These operations require more cognitive resources due to the higher proficiency in L1, resulting in the so-called “**asymmetrical switch costs**” (Bobb & Wodniecka, 2013; Costa et al., 2006; Costa & Santesteban, 2004; Declerck & Koch, 2023). As L2 proficiency increases, this asymmetry between L1 and L2 is expected to diminish.

Asymmetric switch costs were considered indices of inhibition engagement in language control. However, subsequent studies have not consistently supported this hypothesis (e.g., Bobb & Wodniecka, 2013; Costa et al., 2006; Costa & Santesteban, 2004; Declerck & Koch, 2023; Finkbeiner et al., 2006; Titus & Peeters, 2024; Zhu et al., 2022), challenging the ICM and the non-specific language selection hypothesis. For instance, Costa and Santesteban (2004) found asymmetrical switch costs in less proficient bilinguals, but symmetrical switch costs in proficient bilinguals. These results can initially be explained by the non-specific language selection hypothesis, which posits that as L2 proficiency increases, the degree of inhibition between the two languages becomes more balanced, leading to symmetrical switch costs. However, when participants switched between their more proficient L1 and less proficient L2, the switch costs remained symmetrical. They proposed that proficient bilinguals may have developed independent mechanisms for language control. Moreover, some meta-analyses on this topic have not provided solid evidence for the robustness of asymmetric switching costs (e.g., Gade et al., 2021; Goldrick & Gollan, 2023).

Rather than being solely regarded as reflecting the presence or absence of inhibition, asymmetric switch costs are increasingly seen as a continuous measure of reactive language control efficiency (Wu, Ji, et al., 2025). This shift is supported by findings showing that bilinguals with higher language proficiency (e.g., Filippi et al., 2014; Linck et al., 2012) or enhanced general cognitive control (e.g., Liu et al., 2016; Wang et al., 2022) exhibit smaller or even symmetrical switch costs. Thus, instead of delving into the specific mental mechanisms underlying asymmetric switch costs, we focus on the shift from asymmetry to symmetry as a representation of enhanced reactive language control efficiency in the present study.

Inhibition, as previously introduced, is considered as a component of reactive control under the dual control mechanisms framework (Appelbaum et al., 2014; Ma et al., 2016; Seo & Prat, 2019). Accordingly, switch cost is commonly used as an index of reactive language control. In addition to switch cost, mixing cost is another index to assess bilinguals' proactive language control ability. Researchers add single-language blocks to the cued language-switching task (i.e., dual-language blocks). The additional time and errors in non-switch trials

compared to single-language block trials are referred to as **mixing cost** (Declerck, 2020; Declerck, Grainger, & Hartsuiker, 2021; Yang et al., 2024). A greater mixing cost suggests a higher degree of proactive language control. It has been frequently observed that L1 mixing cost is greater than L2 mixing cost, attributed to bilinguals applying greater sustained inhibition to their dominant language L1 in dual-language blocks to optimize performance (e.g., Declerck, 2020; Wang et al., 2022; Wu, Ji, et al., 2024, 2025). Conversely, some researchers propose that asymmetrical mixing costs arise from differing monitoring demands for the two languages (Declerck, 2020). Sustained inhibition and monitoring may overlap, with multiple interpretations of the processes underlying mixing cost in existing studies.

Due to the complexity of material design and variable control, current research primarily focuses on lexical-level language control (referred to as lexical control). However, research suggests that lexical control typically involves the selection and activation of individual words (Costa & Santesteban, 2004; Green, 1998), while language control at other linguistic levels, compared to the lexical level, involves not just the activation of individual lexical items but also their combination and syntactic integration (Chen et al., 2023; Friederici, 2011, 2018; Wu et al., 2019; Zhang & Wang, 2012). Therefore, we assumed that this more complex process may place greater demands on cognitive resources and necessitate adaptations in the language control mechanisms of bilinguals. **Linguistic complexity**, that is, from the lexical level to the super-lexical levels (such as the phrasal level), should serve as a critical dimension to investigate its relationships with language control.

Exploring language control at the super-lexical levels could also shed lights on the question regarding the “**language mapping complexity**” (i.e., to what extent language features of L1 align with those of L2) due to the bilingual typological differences (Ellis, 1989). This further supports the reason why study of language control at the super-lexical levels is of theoretical significance: Linguistic complexity and language mapping complexity induced by bilingual typological differences are suggested to meet and become more salient at the phrasal and even higher linguistic levels (Chen et al., 2023; Su et al., 2024; Tokowicz & MacWhinney, 2005; van Hell & Tokowicz, 2010; Wu, Cheng, et al., 2024). Hence, emerging studies set out to explore language control at the levels of phrases and sentences to evaluate to what extent bilingual typological differences might affect language control (e.g., Chang et al., 2023; Tarlowski et al., 2013; Zhang & Wang, 2012). For instance, an innovative study by Tarlowski et al. (2013) examined Polish-English bilinguals' performance in controlling progressive and perfective phrases. They found symmetrical switch costs for progressive phrases (which are typologically consistent between the two languages) and asymmetrical switch costs for perfective phrases (which are typologically different). This asymmetry suggests that greater typological differences between languages may increase cognitive demands, thereby reducing the efficiency of language control in bilinguals. This highlights bilingual typological differences as a key factor influencing language control. Similarly, Zhang and Wang (2012) examined phrasal-level language control (referred to as phrasal control) in Chinese-English bilinguals, using numeral-classifier-noun phrases with varying bilingual typologies. They found no significant differences in switch costs between phrases like “a glass of water” (which has a structurally similar equivalent in Chinese) and “a cow” (which in Chinese requires the classifier “头/tou2” between “a” and “cow”). Despite these structural differences, their results indicated that bilingual typology did not influence language control.

However, it is noteworthy that Chinese has a much more complex classifier system (see Section 1.2 for details) than English (Bisang, 1999; He, 2001; Her, 2012). According to the hierarchy of difficulty hypothesis for second language acquisition (Ellis, 1989), it is natural to assume that bilinguals with more complex L1 features could easily deal with simpler L2 features. Thus, in Zhang and Wang (2012), Chinese L1 bilinguals did not show the typological effects on language control for the classifier structures. This indicates that in addition to the language typological

differences, the typological difference-induced language mapping complexity might be more critical to influence upon language control, and it is beneficial to take the L1 backgrounds into consideration when examining bilingual phrasal control for constructing a coherent understanding of the nature of language control. Therefore, in this study, we recruited English L1 speakers with much simpler classifier system in their native language to process more complex Chinese L2 numeral-classifier-noun phrases so as to further understand whether and to what extent bilingual typological differences might affect language control when language mapping complexity increases.

To sum up, linguistic complexity should be crucial for investigating the cognitive mechanism underlying language control. Present conclusions drawn from research on lexical control alone may not fully encompass the complexity of language control across different linguistic levels. Moreover, with the increase of linguistic complexity, the language mapping complexity due to bilingual typological differences becomes more salient. Bilinguals with simpler L1 features might resort to more language control demand when processing more complex L2 features. The extent to which previous findings in the field of language control can be generalized requires further empirical validation.

1.2. The complex L2 system: Characteristics of Chinese classifiers

For English-Chinese bilinguals, Chinese L2 has a more complex classifier system as aforementioned. Chinese classifiers are unique compared to Indo-European languages. Structurally, they must appear between numerals and nouns (He, 2001). Functionally, Chinese classifiers serve both classification and measurement purposes (Bisang, 1999; Her, 2012). The uniqueness of Chinese classifiers is in categorizing objects based on inherent and permanent features, such as physical attributes (e.g., “条/tiao2” for long, narrow objects) and function (e.g., “辆/liang4” for vehicles) (Chen, 2021; Zhang, 2007).

Greenberg (1963) compared classifier systems across languages and proposed that plural markers and classifier systems are mutually exclusive, meaning a language cannot have both (Allan, 1977; Li, 2000). English has grammatical number and categorizes nouns as count or non-count based on countability (Her & Hsieh, 2010; Lehrer, 1986). Count nouns denote quantity and combine directly with numerals, while non-count nouns require a quantity word, such as “one glass of water”. According to the Mass Noun Hypothesis (Kuo & Sera, 2009; Lucy, 1996), all nouns in classifier languages like Chinese are considered uncountable. These nouns refer to substances or collectives without clear boundaries rather than bounded objects, making their quantity ambiguous and requiring classifiers to specify it (Greenberg, 2013; Zhang, 2018).

The comprehensive system of Chinese classifiers fosters strong awareness of classifiers among native Chinese speakers. In contrast, Chinese classifiers and classifier-noun collocations pose challenges for native English speakers (Hu, 2012; Lin et al., 2022; Liu & Wang, 2021). The complex correspondences between Chinese and English classifiers cause errors when native English speakers learn Chinese classifiers. According to the hierarchy of difficulty hypothesis, the relationships between Chinese and English classifiers correspond to different levels: Level 0 (no difference between a feature in L1 and L2); Level 1 (two items in L1 become one in L2); Level 2 (one item in L1 is absent in L2); and level 6 (one item in L1 becomes two items in L2) (Ellis, 1989). The error analysis conducted by Lin et al. (2022) found that the most common error type aligns with Level 6, where one item in L1 corresponds to multiple items in L2. Therefore, the richness of Chinese classifiers and their complex correspondences with English pose significant challenges for learners of Chinese as an L2.

Building on Zhang and Wang (2012), this study optimizes participant and material selection. We focus on English L1 speakers learning Chinese as their L2. We categorize English-Chinese classifier-noun phrases into four types: (1) one-to-one correspondence; (2) Chinese-specific classifiers with no English counterparts; (3) one Chinese classifier

corresponds to multiple English classifiers; and (4) one English classifier corresponds to multiple Chinese classifiers (see Materials for details). In line with the hierarchy of difficulty hypothesis (Ellis, 1989), theoretically, for English-Chinese bilinguals, the language mapping complexities of the type of one-to-one correspondence and of the type of one Chinese classifier corresponds to multiple English classifiers should be lower than those of the type of Chinese-specific classifiers with no English counterparts and of the type of one English classifier corresponds to multiple Chinese classifiers, due to the fact that the dominant L1 (English) of these bilinguals is much simpler than their L2 (Chinese) with regard to the classifier system. Therefore, the categorization of these four Chinese L2- English L1 classifier corresponding relationships can systematically and effectively examine how language mapping complexity resulted from bilingual typology differences may affect language control for the present study.

1.3. The relationship between language control and general cognitive control

One perspective posits that language control (control in language domain) may not be a domain-specific mental process but instead rely on general cognitive control (control in non-language-specific cognitive domains) (Blanco-Elorrieta & Caramazza, 2021; De Bruin et al., 2014; Prior & Gollan, 2011; Struys et al., 2019). Investigating the relationship between language control and general cognitive control is essential for addressing whether language control is domain-specific or shares mechanisms with general cognitive control.

The relationship between language control and general cognitive control is often explored by comparing bilinguals' behavioral performance and brain activation patterns in these two domains (e.g., Branzi et al., 2016; Cui et al., 2024; Declerck et al., 2017; Declerck & Grainger, 2017; Jylkkä et al., 2018, 2021; Linck et al., 2012; Liu et al., 2024). Additionally, the training-transfer paradigm has been used to investigate whether training in general cognitive control enhances language control, and vice versa (e.g., Kang et al., 2017; Liu et al., 2019; Prior & Gollan, 2013; Timmer et al., 2019; Wu et al., 2021, 2022).

Some studies have found a significant correlation between language control and general cognitive control, suggesting a shared underlying mechanism (e.g., De Bruin et al., 2014; Kang et al., 2017; Linck et al., 2012; Liu et al., 2014, 2016; Struys et al., 2019; Timmer et al., 2019). Specifically, Liu et al. (2014) found that bilinguals with lower general cognitive control had asymmetrical switch costs, while those with higher general cognitive control showed symmetrical switch costs. This indicates that general cognitive control influences bilinguals' language control. However, some studies suggest that language control may be domain-specific (e.g., Branzi et al., 2016; Calabria et al., 2012, 2015; Heikoop et al., 2016; Stassen et al., 2017; Wolna et al., 2024). Calabria et al. (2012) hypothesized that if these two domains share the same mechanisms, the control patterns should be consistent. However, they found that bilinguals exhibited symmetrical switch costs in language control but asymmetrical switch costs in general cognitive control, suggesting a fundamental distinction between language control and general cognitive control.

According to the dual control mechanisms framework, individuals utilize both proactive and reactive control depending on their control demands (Braver, 2012; Braver et al., 2009). The selection between these two control mechanisms is influenced by external factors and individual differences (Burgess et al., 2010; Burgess & Braver, 2010; Green & Abutalebi, 2013). The aforementioned studies have focused on reactive control, neglecting the adaptive adjustments within bilinguals' control systems.

However, some studies have examined factors that modulate the relationships (e.g., Jiang et al., 2023, 2024; Li et al., 2021; Liu et al., 2019; Wang et al., 2022). Wang et al. (2022) found that as L2 proficiency increases, bilinguals' language control mechanisms shift from relying on reactive control to proactive control. Jiang et al. (2023) investigated

bilinguals' proactive and reactive language control in alternation and dense code-switching contexts, and their relationships with general cognitive control. Results revealed that general reactive cognitive control is engaged in the alternation context, while general proactive cognitive control is engaged in the dense code-switching context. These findings indicate that different contexts impose distinct control demands for bilinguals, which subsequently influence the relationship between language control and general cognitive control.

Hence, bilinguals may adjust their control mechanisms in response to various language control demands, which may influence the relationships between language control and general cognitive control. The controversial findings in prior studies may arise from overlooking the dynamic cognitive mechanisms underlying language control. For instance, research suggests that bilinguals with high L2 proficiency tend to rely more on proactive control (Wang et al., 2022). In this context, examining the relationship between language control and general cognitive control solely from a reactive control perspective may be somewhat inevitable to result in conflicting findings and be insufficient to capture the full complexity of the dynamic cognitive mechanisms involved in multi-level language control.

In summary, bilinguals' control mechanisms might be dynamic and adaptively adjusted with regards to the control demands. We hypothesize that language control may share mechanisms with general cognitive control but would also be moderated by external factors (e.g., context) and individual differences (Jiang et al., 2023, 2024; Wang et al., 2022). While previous research has emphasized the role of external factors in the dynamic nature of language control, our current study aims to examine this relationship from the perspective of internal language factors (e.g., linguistic levels), particularly across lexical and phrasal levels, in a more language-focused manner.

1.4. The current study

Research has primarily focused on reactive lexical control and its relationship with general cognitive control (e.g., Costa et al., 2006; De Bruin et al., 2014; Schwieter & Sunderman, 2008; Timmer et al., 2019; Wu et al., 2021, 2022). However, these studies did not focus on the linguistic complexity and the dynamic adaptive adjustments bilinguals make under varying control demands. Given that linguistic complexity represented by different linguistic levels might lead to various control demands, this study examined bilinguals' proactive and reactive language control at both lexical and phrasal levels, and their relationships with general cognitive control.

Following established methodologies, we utilized nouns as materials for lexical control. Additionally, the typology differences between English and Chinese classifiers, and the complexity of Chinese classifiers for English L2 learners justified the numeral-classifier-noun phrases as ideal materials for examining phrasal control.

To measure bilinguals' language control abilities, this study employed the cued language-switching paradigm (Meuter & Allport, 1999) to guarantee the current findings to be comparable with those of related studies. Mixing costs and switch costs were used as indices to measure proactive and reactive language control abilities, respectively.

The AX-CPT task (Braver et al., 2005; Wiemers & Redick, 2018) and the Stroop task (Bialystok et al., 2008; Moretti et al., 2025; Stroop, 1935) were used to assess general proactive and reactive cognitive control abilities, respectively. Among tasks measuring general reactive cognitive control, the Stroop task (Bialystok et al., 2014; Moretti et al., 2025) and the Simon task (Bialystok et al., 2005; Martin-Rhee & Bialystok, 2008) are commonly employed. Both the Stroop and Simon tasks involve interference resolution but differ in the nature of conflict. The Simon task primarily taps stimulus-response (S-R) conflict, where the stimulus (e.g., location) does not match the response rule (e.g., pressing the left key when the stimulus appears on the right) (Blumenfeld & Marian, 2014; Kornblum et al., 1999). In contrast, the Stroop task involves both stimulus-stimulus (S-S) and S-R conflict: the word meaning (e.g., red)

conflicts with its displayed color (e.g., blue), resulting in S-S interference, while the irrelevant stimulus feature has dimensional overlap with the response set, eliciting S-R conflict (Kornblum et al., 1999). Despite differences in stimulus types, these tasks are often used to assess inhibition (general reactive cognitive control) because both require individuals to focus on one aspect of the stimulus while ignoring a conflicting feature (Hui et al., 2020). However, some studies suggest that bilinguals may experience co-activation of both languages, leading to more frequent S-S conflict. (Bialystok et al., 2008; Blumenfeld & Marian, 2014; Costa et al., 2009; Hernández et al., 2010). Therefore, the Stroop task, by engaging both S-S and S-R inhibition, offers a theoretically grounded measure of general reactive cognitive control, which is particularly relevant to bilingualism. Based on these considerations, we selected the Stroop task as our synthetic measure of general reactive cognitive control.

To sum up, the core research question of this study is: What cognitive mechanisms underlie bilinguals' lexical and phrasal control, and how do they differ? This question is subdivided into three main research questions:

Research Question 1. What are bilinguals' behavioral performances in proactive and reactive lexical control?

Research Question 2. What are bilinguals' behavioral performances in proactive and reactive phrasal control?

Research Question 3. What are the relationships between lexical control, phrasal control, and general cognitive control in bilinguals?

The relationships among these questions are as follows: based on Braver's (2012) dual control mechanisms framework, we will examine bilinguals' control mechanisms from both proactive and reactive control dimension. First, we aim to identify specific patterns of language control in bilinguals at both the lexical and phrasal levels (Research Questions 1 and 2). Next, we will explore the relationship between language control and general cognitive control to uncover the underlying cognitive mechanisms (Research Question 3).

The hypotheses for the research questions are as follows:

Hypothesis 1. According to the inhibitory control model and the Non-Specific Language Selection Hypothesis (Green, 1998; Meuter & Allport, 1999), the degree of reactive language control is linked to language proficiency. For most bilinguals, L1 proficiency significantly exceeds that of L2, resulting in higher reactive control demands for L1 and, consequently, greater switch costs for L1 than for L2. As L2 proficiency increases, the difference in reactive control demands between L1 and L2 diminishes, leading to more symmetric switch costs. Given the high L2 proficiency of bilinguals in this study, the reactive language control demands for lexical control should be comparable for both languages.

Additionally, bilinguals are likely to engage stronger proactive control mechanisms to consistently monitor or inhibit the dominant language (L1), thereby balancing activation between languages to optimize overall performance, as demonstrated in previous studies (e.g., Declerck, 2020; Wu et al., 2021; Wu, Ji, et al., 2025). Consequently, we expect higher mixing costs for L1 than for L2.

Hypothesis 2. Due to linguistic complexity, the demands for reactive phrasal control are expected to exceed those for reactive lexical control, resulting in reduced efficiency in reactive language control. This should be reflected in asymmetric switch costs between L1 and L2, even among proficient bilinguals in the current study.

Furthermore, if language mapping complexity aroused by the bilingual typological differences would affect language control, significant differences in switch costs are expected across the four types of classifier-noun phrases. Based on the hierarchy of difficulty hypothesis (Ellis, 1989), switch costs for Type 1 phrases (one-to-one correspondence) and Type 3 phrases (one Chinese classifier corresponding to multiple English classifiers) should be significantly smaller than those

for Type 2 phrases (Chinese-specific classifiers with no English counterparts) and Type 4 phrases (one English classifier corresponding to multiple Chinese classifiers).

Unlike reactive control, which is more sensitive to bottom-up, stimulus-related features (Braver, 2012; Seo & Prat, 2019), proactive control is more general and related to the dominance of the target language (Braver, 2012; Zhu & Sowman, 2020). Given that English remains the dominant language for the bilinguals in this study, we predict greater L1 than L2 mixing costs at the phrasal level, consistent with Hypothesis 1.

Hypothesis 3. Language control is expected to depend on general cognitive control, though this relationship is dynamic. Previous studies have shown that proficient bilinguals rely more on proactive than reactive cognitive control for lexical control (Wang et al., 2022). This aligns with Hypothesis 1, which predicts that the reactive control demands of proficient bilinguals will balance between the two languages, leading to improved reactive control efficiency. Therefore, we hypothesize that bilinguals will primarily rely on general proactive cognitive control for lexical control. In contrast, as predicted by Hypothesis 2, the reactive control demands for bilinguals are expected to increase during phrasal control. Thus, bilinguals may depend more on general reactive cognitive control for phrasal control.

2. Methods

2.1. Participants

Thirty-three English-Chinese bilinguals (20 females; average age = 23.25 years; $SD = 3.60$ years) participated in this study. Since adjustments in trial duration for the formal experiment, six participants from the pilot study were excluded. Due to the experiment materials for both lexical control and phrasal control tasks were composed of intermediate-level contents, participants who had passed HSK (i.e., Hanyu Shuiping Kaoshi) Band-4, an intermediate level on the standard Chinese L2 proficiency test (ranging from Band-1 for beginners to Band-6 for advanced learners) were recruited.

Moreover, to further screen the suitable participants, we used the Chinese L2 proficiency quick test developed by Feng et al. (2020), which required participants to complete 30 fill-in-the-blank items within 15 min. Errors were scored as 0 points, with a maximum score of 30 points. Participants scoring above 18 points were considered to have advanced proficiency (Feng et al., 2020, 2023; Zhou & Jiang, 2023). Three participants scoring below 18 were excluded to ensure that all participants were relatively proficient L2 learners. The average Chinese L2 proficiency score of the remaining participants was 22.83 ($SD = 4.31$).

Twenty-four participants (14 females; average age = 22.63 years; $SD = 3.03$ years) were included in the final data analysis, comprising 10 Americans, 12 British, and 2 Canadians. Among the participants, sixteen spoke one or more additional languages, including Bangla ($n = 2$), Italian ($n = 2$), Japanese ($n = 2$), German ($n = 1$), Bahasa Indonesia ($n = 4$), Sinhala ($n = 2$), French ($n = 2$), Russian ($n = 1$), and Turkish ($n = 1$). Participants began learning English from birth and started learning Chinese at an average age of 12.72 years ($SD = 7.36$), qualifying them as late Chinese L2 learners (Festman et al., 2010; Weber-Fox & Neville, 1996).

Self-assessed proficiency in L1 and L2 was measured using a 4-point scale across four dimensions: listening, speaking, reading, and writing (1 = Scarcely proficient, 4 = Perfectly proficient), with a maximum score of 16. Higher scores reflect greater self-assessment of language proficiency. The mean self-rating for L1 proficiency was 14.7 ($SD = 1.55$), and the mean self-rating for L2 proficiency was 11.79 ($SD = 1.10$).

Language switching frequency was assessed using the Bilingual Switching Questionnaire (BSWQ) (Rodríguez-Fornells et al., 2012). Participants rated their language switching frequency on a 5-point scale (1 = never, 5 = always). The questionnaire comprised four dimensions:

(1) L1 Switch (assessing the tendency to switch from L2 to L1); (2) L2 Switch (assessing the tendency to switch from L1 to L2); (3) Contextual Switch (assessing language switching in specific contexts or situations); and (4) Unintended Switch (assessing unconscious language switching). Each dimension consisted of 3 items, with a maximum score of 15 per dimension, resulting in a total possible score of 60. Participants reported an average total score of 36.95 ($SD = 5.55$), with sub-scores of 11.08 ($SD = 2.36$) for L1 Switch, 7.29 ($SD = 2.07$) for L2 Switch, 10.25 ($SD = 2.82$) for Contextual Switch, and 8.42 ($SD = 1.67$) for Unintended Switch.

All participants were right-handed, as determined by the Edinburgh Handedness Inventory (Oldfield, 1971). They had understood the experimental requirements and signed the “Experiment Informed Consent”. After experiment, each participant received 200 CNY, with an additional 20 CNY reward for excellent compliance.

2.2. Materials

We examined the lexical and phrasal control performances in English-Chinese bilinguals employing the cued language-switching paradigm based on picture-naming.

The experimental materials included concrete nouns for the lexical control task, numeral-classifier-noun phrases for the phrasal control task, and their corresponding pictures. Detailed materials were presented in Table S1–S3 in the *Supplementary Materials*.

2.2.1. Noun materials

All nouns were selected from the *Chinese Proficiency Grading Standards for International Chinese Language Education* (2021). The same set of nouns was used in both the lexical control and phrasal control tasks, where the nouns in the phrasal control task were incorporated into numeral-classifier-noun phrases. The difficulty of the noun materials was matched with the participants' Chinese L2 proficiency, and, therefore, participants had already mastered these nouns in accordance with the course syllabus before the actual experiment. Participants would be required to reacquaint themselves with the noun materials to minimize the potential familiarity effect (see also Section 2.3).

2.2.2. Phrase materials

The phrase materials were categorized into four types based on the relationships between English and Chinese classifiers (Ellis, 1989):

- (1) Type One: one-to-one correspondence (e.g., “杯/bei1” corresponds to “glass”).
- (2) Type Two: Chinese-specific classifiers with no English counterparts (e.g., “颗/ke1” with no corresponding English classifier).
- (3) Type Three: one Chinese classifier corresponds to multiple English classifiers (e.g., “串/chuan4” corresponds to “string” and “bunch”).
- (4) Type Four: one English classifier corresponds to multiple Chinese classifiers (e.g., “pair” corresponds to “双/shuang1,” “副/fu4,” and “条/tiao2”).

The design process involved selecting appropriate classifiers, controlling for the typicality and frequency of classifier-noun collocations, and assessing their semantic acceptability.

- (1) **Selection of classifiers:** we selected classifiers from the *Chinese Proficiency Grading Standards for International Chinese Language Education* (2021), covering from Band 1 (primary proficiency) to 9 (advanced proficiency). To ensure the feasibility of picture naming, we selected classifiers that collocated with semantically concrete nouns and excluded those from Bands 7 to 9 to match the Chinese L2 proficiency of the participants. According to the teaching schedule, these classifiers should have been learned already before the actual experiment. In order to exclude the familiarity effect and get the participants adapted themselves

with the experimental materials, they would receive the phrase materials before the experiment to get re-familiarized with the classifiers as well as the corresponding phrases they had formerly acquired (see also Section 2.3).

- (2) **Controlling of classifier-noun collocation typicality:** we conducted an initial screening of classifiers with various bilingual correspondences based on *A Chinese-English Dictionary of Measure Words* (2001) and *A Dictionary of Chinese Classifiers with English Equivalents* (2002), and referred to definitions in the *Oxford Advanced Learner's English-Chinese Dictionary* (2008). To reduce the subjectivity inherent in dictionary-based selection, we adopted the Corpus of Contemporary American English (COCA) (<https://www.english-corpora.org/coca/>) and the Beijing Language and Culture University Corpus (BCC) (<http://bcc.blcu.edu.cn/>). We first excluded classifiers with inconsistent correspondences between dictionaries and corpora. Next, we excluded classifiers with limited collocations that typically match only specific nouns. Finally, among classifiers with similar semantics or phonetics but unbalanced collocations, we selected those with more general collocations.
- (3) **Controlling of classifier-noun collocation frequency:** We searched COCA and BCC to select high-frequency Chinese-English classifier-noun collocations, and excluded nouns that could collocate with multiple semantically similar classifiers or had high collocation frequencies in the Chinese corpus but no corresponding collocations in English. The frequency of a classifier C with all possible nouns was F1, and its frequency with specific nouns was F2. The collocation frequency was calculated as F2/F1. This process resulted in the final selection of 20 Chinese classifiers and 14 English classifiers. Based on high-frequency collocations, we compiled 106 Chinese classifier-noun phrases and 106 English classifier-noun phrases for further analysis.
- (4) **Assessment of the semantic acceptability of classifier-noun phrases:** 12 native Chinese speakers and 12 native English speakers, who did not participate in the formal experiment, respectively rated the acceptability of 106 Chinese phrases and 106 English phrases using a 5-point scale (1 = extremely unacceptable and 5 = fully acceptable). Based on these ratings, we selected a total of 90 phrases (45 in each language) with scores of 4.5 or higher as the materials. All numerals in phrases were standardized to “一/yi1” in Chinese and “one” in English to ensure the numeral consistency and to minimize the naming effects from the numerals.

To address the potential impacts of material types on participants, we further analyzed the acceptability results of the four types of phrases. Friedman tests showed no significant differences among the four types ($\chi^2(3) = 5.482, p = 0.140$). Similarly, for English classifier-noun phrases, Friedman tests indicated no significant differences ($\chi^2(3) = 1.93, p = 0.588$).

Each type of numeral-classifier-noun phrase included 12 Chinese phrases and 12 English phrases, with each classifier collocated with two nouns (45 Chinese nouns and 45 English nouns). Nevertheless, in Type Three and Type Four, three English classifiers “string,” “piece” (correspond to Chinese classifier “张/zhang1”), and “pair” (corresponds to Chinese classifier “条/tiao2”) could not be matched with two nouns each. Therefore, these phrases were repeated to balance the number of materials. In the formal experiment, the sequence of all trials was pseudo-randomized, ensuring that the same trial appeared only after an interval of at least four trials.

2.2.3. Picture materials

We prepared 45 color pictures sourced from online searches and the psycholinguistic picture naming database (Brodeur et al., 2014). Photoshop CC 2019 was utilized to standardize the format and visual complexity of these pictures. To ensure consistency in picture naming,

three native Chinese speakers and three native English speakers, respectively reviewed the pictures. They received an online questionnaire that included each picture accompanied by a corresponding noun or phrase, and two choices provided. A “1” indicates that the picture and the content match, while a “0” indicates a mismatch. If a picture was marked as “0,” an interview with the participant was conducted, and adjustments were made based on their feedback. In the formal experiment, participants were required to name the pictures according to the

provided examples. The same pictures were employed for both lexical control and phrasal control tasks. For the phrasal control task, we increased the number of animal pictures corresponding to the Type Three numeral-classifier-noun phrases and the pairs of objects for the Type Four numeral-classifier-noun phrases to match the picture-phrase meanings.

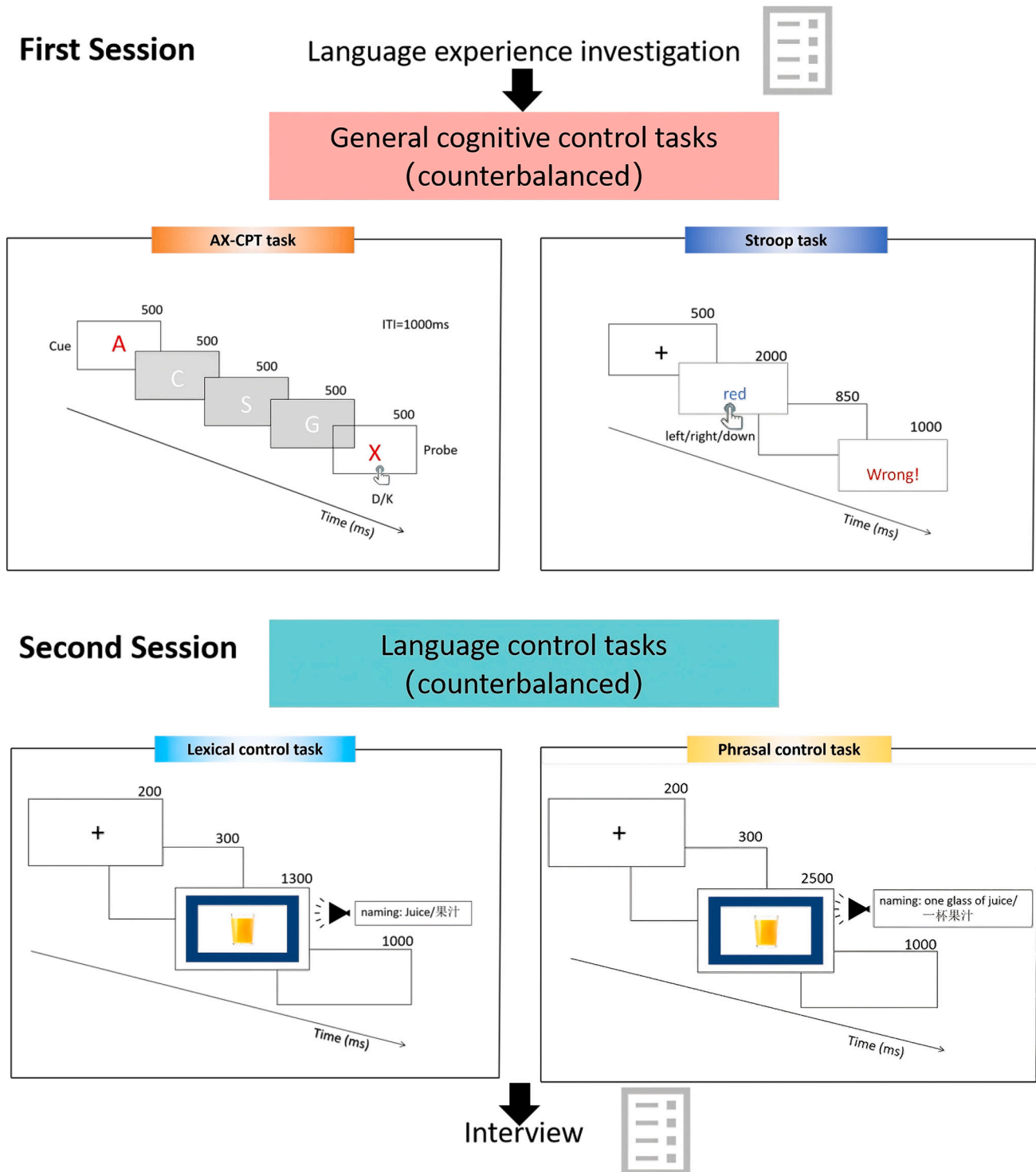


Fig. 1. Experiment procedure. In the first session, participants completed a language experience investigation, followed by two general cognitive control tasks (AX-CPT and Stroop tasks) in a counterbalanced order. The second session involved two language control tasks (Lexical and Phrasal control tasks), also presented in a counterbalanced order. After completing all tasks, participants were interviewed. The two sessions were spaced 3 to 5 days apart.

2.3. Procedure

The formal experiment consisted of two sessions. In the first session, participants were instructed to complete the language experience investigation, and two general cognitive control tasks (the AX-CPT task and the Stroop task). In the second session, participants completed the lexical and phrasal control tasks. The sequence of tasks was counterbalanced across participants.

The two sessions were 3 to 5 days apart, with a 15-min resting time between each task. After the experiment, participants were interviewed to gain a more detailed understanding of their mental processes during the tasks (Fig. 1). Experimental stimuli were presented using the Psychopy version 2022.2.4 (<https://www.psychopy.org/>).

2.3.1. The lexical control task

The lexical materials were sent to participants three to five days prior to the formal experiment to ensure they had enough time to re-familiarize themselves (Wang et al., 2022; Wu et al., 2022; Wu, Ji, et al., 2024; Yang et al., 2024), minimizing potential effects of participants' familiarity on the results. Participants received materials that included Chinese nouns with pinyin, English nouns, English-Chinese bilingual audio recordings, and corresponding pictures. They were instructed to re-familiarize themselves with the nouns and their corresponding pictures, with the audio recordings provided as a standard pronunciation reference. A pre-test was conducted before the formal experiment, and only participants who achieved an average accuracy (ACC) of 90 % or higher were deemed to master the lexical knowledge and thus allowed to proceed. All participants had met this criterion.

In the formal experiment, participants were instructed to use the correct language based on the color of the frame (red or blue, as the cues) to name pictures quickly and accurately. Each trial began with a fixation point (“+”) at the center of the screen for 200 ms, followed by a 300 ms blank screen. A picture framed in either red or blue was then presented for 1300 ms, followed by a 1000 ms blank screen between trials (Fig. 1). Each trial had a total duration of 2800 ms (Wang et al., 2022; Wu et al., 2021), and the entire task lasted for 18 min.

The formal experiment consisted of single-language and dual-language blocks. In the single-language block, participants named pictures in only one language (L1 or L2). In the dual-language block, participants named pictures in either L1 or L2 according to cues, resulting in switch trials (where a different language is required compared to the preceding trial, i.e., L1 → L2, L2 → L1) and non-switch trials (where the same language is required as in the preceding trial, i.e., L1 → L1, L2 → L2). To prevent expectation effects, the “sandwich” method was used to counterbalance the language blocks (Declerck, 2020; Wang et al., 2022; Wu, Ji, et al., 2024, 2025). This method involved three parts: first, presenting single-language blocks for L1 and L2; second, performing the dual-language block; and finally, presenting the single-language blocks again. The language blocks and cues were counterbalanced across participants.

The trial sequences in single-language blocks were randomly ordered. In the dual-language block, the language type (L1 or L2), trial type (switch or non-switch), and picture stimuli were pseudo-randomized utilizing MATLAB R2021a (<http://www.mathworks.cn/>). The same language appeared a maximum of two consecutive trials, the same trial type a maximum of four consecutive times, and the same picture stimulus could reappear only after at least four trials. Five lists were generated for both single-language and dual-language blocks, with the experiment program randomly selecting one set of these lists.

The formal experiment comprised two single-language blocks for L1, two for L2, and one dual-language block, totaling 389 trials. Each block began with a filler trial, which was excluded from the final data analysis (Costa et al., 2006; Costa & Santesteban, 2004; Meuter & Allport, 1999; Wang et al., 2022). Each single-language block included 48 formal trials, while the dual-language block contained 192 formal trials. Within the dual-language block, there were 48 trials each for L1 non-switch, L1

switch, L2 non-switch, and L2 switch conditions.

2.3.2. The phrasal control task

Similarly, phrase materials were sent to the participants together with the lexical materials three-to-five days before the actual experiment. This was to reacquaint themselves with the classifiers as well as the phrases which had been learned already. The mastery of the phrases was tested using the same procedure and standards as those used in the lexical control task. In the formal experiment, participants were required to select the correct language based on the color of the frame to name the pictures in phrases quickly and accurately. Each trial began with a fixation point (“+”) at the center of the screen for 200 ms, followed by a blank screen for 300 ms. Then, a picture with either a red or blue frame was displayed for 2500 ms. A blank screen was shown for 1000 ms between trials (Fig. 1). Each trial lasted a total duration of 4000 ms, and the entire task lasted for 25 min.

The formal experiment comprised two single-language blocks for L1, two for L2, and one dual-language block. Each single-language block included 48 trials, while the dual-language block contained 192 trials. Within the dual-language block, there were 48 trials each for L1 non-switch, L1 switch, L2 non-switch, and L2 switch conditions. Additionally, the phrase materials were categorized into four types, with each type containing 12 trials for L1 non-switch, L1 switch, L2 non-switch, and L2 switch conditions. The methods for balancing the cues, language blocks, and trial sequences were consistent with those used in the lexical control task.

2.3.3. The AX-CPT task

The AX-CPT task was used to assess participants' general proactive cognitive control. In this task, sequences of capital letters were presented (in a black background), with each trial consisting of five letters displayed in the following sequence and colors: red, white, white, white, and red. The first red letter served as the cue, the last red letter as the probe, and the three white letters in between acted as fillers (Gonthier et al., 2016; Wiemers & Redick, 2018).

The AX-CPT task included four types of trials: AX, AY, BX, and BY. Only the AX trials were the target trials. At the probe onset, participants were required to press one of two response keys, associated with either target or non-target responses. If X occurred after A, the target button (key “D”) was pressed. For any other sequence of stimuli (AY, BX, BY), the non-target button (key “K”) was pressed (Braver et al., 2005; Kubota et al., 2020; Wiemers & Redick, 2018).

The formal experiment consisted of 100 trials in total, including 70 AX trials and 10 trials each for BX, BY, and AY. The fillers consisted of 20 English letters other than “A,” “X,” “B,” “Y,” “D,” and “K,” and their sequence was balanced. Each trial began with the cue displayed at the center of the screen for 500 ms, followed by the three fillers, each shown for 500 ms. The probe was then presented for 500 ms. A blank screen was shown for 1000 ms between each letter (Fig. 1). Each trial lasted a total of 7500 ms (see also Wiemers & Redick, 2018), and the entire task lasted for 15 min.

2.3.4. The Stroop task

Participants completed the Stroop task to assess their general reactive cognitive control abilities. They were instructed to ignore the word semantics and to respond only to the colors by pressing the left key for red, the right key for green, and the down key for blue (Soutschek et al., 2013). Three different English color words were presented (“red,” “green,” and “blue”), creating both congruent and incongruent conditions. In the congruent condition, the semantics of the words matched the color (e.g., the word “red” displayed in red). In the incongruent condition, the semantics of the words conflicted with the color (e.g., the word “red” displayed in green).

The formal experiment consisted of 120 trials, equally divided between congruent and incongruent conditions. Each trial began with a fixation point (“+”) at the center of the screen for 500 ms, followed by

the stimulus for 2000 ms, and then a blank screen for 850 ms. Participants received feedback for 1000 ms after responding to the stimulus (Fig. 1). Each trial lasted a total of 4350 ms (Wu et al., 2022), and the entire task lasted for 10 min.

2.4. Data analysis

Before analyzing the data from both the lexical and phrasal control tasks, we utilized Praat version 6.3.1.7 (<https://praat.en.softonic.com>) for noise reduction and transcription in each trial. RT and ACC for each trial were extracted and error types were categorized. We discarded the following types of trials: trials where responses were given in an incorrect language or word, trials involving coughing or pausing, trials following a response in an incorrect language, and trials with recording failures or absolute outliers (RTs below 500 ms or above 2000 ms for the lexical control task, and below 500 ms or above 4000 ms for the phrasal control task).

In the AX-CPT task, the Proactive Behavioral Index (PBI) was used to measure participants' general proactive cognitive control abilities. The PBI is calculated as $(AY - BX) / (AY + BX)$ for RTs, which assesses the relative balance of interference between AY and BX trials. A more positive PBI indicates higher interference in AY trials, suggesting a general proactive cognitive control tendency (Kubota et al., 2020). The participants' PBI was 0.12 ($SD = -0.16$). For the Stroop task, general reactive cognitive control abilities were assessed utilizing the Stroop score, derived from the difference in RTs between congruent and incongruent trials. A smaller Stroop score indicates greater general reactive cognitive control abilities (Bialystok et al., 2008; Soutschek et al., 2013; Stroop, 1935). The participants' Stroop score was 60 ms ($SD = 8$). Moreover, some studies suggest that the Stroop task may involve sequential effects, where the congruency effect is often reversed following incongruent trials (Gratton et al., 1992; Moretti et al., 2025). In light of this, we also computed the Stroop scores for bilinguals based on the n-1 congruent trials. The new Stroop score was 92 ms ($SD = 11$).

In this study, we primarily employed the linear mixed-effects model (LMEMs) using the lme4 package (Bates et al., 2015) and the lmerTest package (Kuznetsova et al., 2017) to analysis the reaction time (RT) data. We employed LMEMs instead of traditional ANOVA for the following methodological reasons. First, LMEMs allow simultaneous incorporation of both fixed effects (e.g., experimental manipulations and control variables) and random effects (e.g., participant- and item-specific variations), thereby unifying the conventional by-participant (F1) and by-item (F2) analyses within a unified framework (Baayen et al., 2008; Wu, Ji, et al., 2025). This integration improves model efficiency while explicitly accounting for dependencies in clustered data, avoiding the strict independence assumption required by traditional regression (Bates et al., 2015). Second, through partial pooling techniques (Gelman & Hill, 2006), LMEMs demonstrate relative robustness in small-sample scenarios, particularly when the random-effects structure is appropriately simplified (Matuschek et al., 2017; McManus, 2021). By estimating variance components for both participants and items, the models reduce bias in fixed-effect estimates compared to ANOVA approaches that treat these factors as fixed (Barr et al., 2013; Jiang et al., 2023; McManus, 2021). Third, due to the surge of employing LMEMs for data analysis in the field of language control (e.g., Wang et al., 2022; Wu, Ji, et al., 2024; Wu, Zhao, et al., 2025; Yang et al., 2024), we hope that on the basis of LMEMs, the findings of this study could be comparable with the recent results.

We initially constructed a full model with the maximum random effects structure, including random intercepts and random slopes for both participants and items (Barr et al., 2013). If the models failed to converge, we used a backward-stepping procedure to refine it, selecting the best-fitting model through model comparisons (Barr et al., 2013; McManus, 2021). We employed Restricted Maximum Likelihood (REML) for estimation and applied ANOVA to reduce the random intercepts in the full model (Barr et al., 2013; McManus, 2021). The

decision to include random slope effects was based on Akaike Information Criterion (AIC) comparisons, with the model exhibiting the smallest AIC value identified as the best-fitting model (Barr et al., 2013; Zhou & Jiang, 2023). The significance level for rejecting the null hypothesis was set at $\alpha = 0.05$.

To analyze the error rate (ER) data, we used a logistic mixed-effects model with the glmer function (Baayen et al., 2008). However, the ER data were not the primary focus of our study for the following reasons:

- (1) No tradeoff between RTs and ERs. For each participant, we computed the correlation between mean RTs and mean ERs across conditions. In the lexical control task, the average correlation was positive ($r = 0.30$), and a one-sample t -test confirmed that it was significantly greater than zero, $t(23) = 3.52, p = 0.002$ (two-tailed). In the phrasal control task, the average correlation was also positive ($r = 0.08$), but not statistically significant, $t(23) = 0.80, p = 0.43$ (two-tailed). These results provide no evidence of a systematic negative correlation between speed and accuracy, indicating no speed-accuracy tradeoff.
- (2) Low ER values: The ERs were relatively low, as shown in the descriptive statistics in Sections 3.1 and 3.2. This suggests that ERs did not present substantial variation across conditions, making them less informative for the current study.
- (3) Low sensitivity of ERs: Prior research on language control has demonstrated that ERs are less sensitive compared to RTs (Declerck et al., 2019; Wu, Zhao, et al., 2025). Moreover, no switch costs were observed in the ER data of the current study, despite their being a key index in language control research. These findings align with prior results (e.g., Declerck et al., 2019; Declerck & Grainger, 2017; Jackson et al., 2004; Struys et al., 2019; Von Studnitz & Green, 2002; Wu, Zhao, et al., 2025), reinforcing our decision to prioritize RTs as the primary measure in our analysis.

In summary, ERs are not a reliable index for the current study. We focus primarily on RT data for analysis and discussion. The detailed ER results are provided in Table S4–S7 in the *Supplementary Materials*.

The full model was constructed based on mixing costs (proactive language control index) and switch costs (reactive language control index). Mixing costs were determined by comparing RTs between non-switch trials in the dual-language block and trials in the single-language block (Declerck, 2020). Switch costs were determined by comparing RTs between switch trials and non-switch trials in the dual-language block (Declerck & Philipp, 2015a). The data analysis processes were as follows:

- (1) For the analysis of both lexical and phrasal control (Research Questions 1 and 2), the fixed effects of proactive language control were: Language (L1, L2), Trial type (single-language, non-switch), and their interaction. The fixed effects of reactive language control were: Language (L1, L2), Trial type (non-switch, switch), and their interaction. Additionally, based on the normality of switch costs across the four classifier-noun phrase types, we performed either a one-way ANOVA or a Friedman test. Given that proactive language control entails sustained, overall regulation of language, we did not compute mixing costs separately for each phrase type.
- (2) For Research Question 3, we examined the relationship between language control at different linguistic levels and general cognitive control using both mixed models and Spearman correlation analysis. In the mixed model, we incorporated general cognitive control index (Stroop score and PBI) into separate models for lexical and phrasal control, following established methodologies (e.g., Bisson, 2023; Farqi-Shah et al., 2021; Jiang et al., 2023; Momenian et al., 2024; Wang et al., 2022; Yi, 2022). We focused on the interactions between the general cognitive control index

(i.e., PBI and Stroop score) and trial type, as well as between the general cognitive control index, language, and trial type. The first analysis examined whether general cognitive control predicts language control, while the second analysis examined whether general cognitive control influences language control efficiency.

Additionally, while Spearman correlations provided initial insights, their limitations in handling repeated measures and unobserved heterogeneity made them less suitable for our data. In contrast, the mixed-effects framework offered three key advantages: (1) explicit modeling of within-participant dependencies using random intercepts, (2) simultaneous control of confounding variables through fixed-effect covariates, and (3) greater robustness against Type I errors in small samples compared to zero-order correlations (Barr et al., 2013; Bates et al., 2015; Jiang et al., 2023; McManus, 2021). Given these advantages, we prioritize the mixed-model results in the main analysis, with the full correlation results are provided in Appendix S2 in the *Supplementary Materials* for supplementary information.

Log-transformed RTs were used as the dependent variable, with participants and items treated as random effects. Trial type and Language were contrast coded (L1 = -0.5 and L2 = 0.5; non-switch = 0.5 and single-language = -0.5; switch = 0.5 and non-switch = -0.5). PBI and Stroop scores were treated as continuous predictors and centered. In addition, we incorporated L2 proficiency score and language switching frequency as control variables in these models to exclude the interference of irrelevant factors. These control variables were centered at the sample mean in the model (Jiang et al., 2023, 2024; Zhou & Jiang, 2023).

All analyses were conducted with R version 4.2.3 (<https://cloud.r-project.org>). The R models were provided in Appendix S1 of the *Supplementary Materials*.

3. Results

3.1. Lexical control

The mean RTs and ERs for the lexical control task are presented in Table 1.

In the analysis of mixing costs, the main effect of Trial type was significant ($b = 0.063$, $SE = 0.010$, 95 % $CI = [0.043, 0.083]$, $t = 6.43$, $p < 0.001$), indicating that RTs in non-switch trials (1091 ms) were longer than in single-language trials (1027 ms), resulting in an averaged mixing cost of 64 ms. The main effect of Language was not significant ($b = 0.001$, $SE = 0.020$, 95 % $CI = [-0.041, 0.043]$, $t = 0.05$, $p = 0.961$), suggesting that naming speed for L1 (1044 ms) did not differ from that for L2 (1052 ms). Additionally, the interaction between Trial type and Language was significant ($b = -0.041$, $SE = 0.010$, 95 % $CI = [-0.060, -0.021]$, $t = -4.07$, $p < 0.001$), revealing an asymmetry in mixing costs, with 83 ms for L1 and 45 ms for L2.

In the analysis of switch costs, the main effect of Trial type was significant ($b = 0.094$, $SE = 0.006$, 95 % $CI = [0.082, 0.105]$, $t = 15.95$, $p < 0.001$), indicating that RTs were longer for switch trials (1199 ms)

compared to non-switch trials (1091 ms), resulting in an averaged switch cost of 108 ms. The main effect of Language was not significant ($b = -0.023$, $SE = 0.019$, 95 % $CI = [-0.062, 0.016]$, $t = -1.22$, $p = 0.234$), showing no difference in naming speed between L1 (1158 ms) and L2 (1132 ms). Additionally, the interaction between Trial type and Language was not significant ($b = -0.010$, $SE = 0.012$, 95 % $CI = [-0.033, 0.013]$, $t = -0.82$, $p = 0.415$), revealing symmetrical switch costs of 115 ms for L1 and 102 ms for L2.

3.2. Phrasal control

The mean RTs and ERs for the phrasal control task are presented in Table 2.

In the analysis of mixing costs, the main effect of Trial type was significant ($b = 0.067$, $SE = 0.016$, 95 % $CI = [0.034, 0.100]$, $t = 4.19$, $p < 0.001$), indicating that RTs were longer for non-switch trials (1273 ms) compared to single-language trials (1194 ms), resulting in an averaged mixing cost of 79 ms. The main effect of Language was not significant ($b = 0.007$, $SE = 0.017$, 95 % $CI = [-0.029, 0.043]$, $t = 0.40$, $p = 0.696$), suggesting no difference in naming speed between L1 (1210 ms) and L2 (1231 ms). Additionally, the interaction between Trial type and Language was significant ($b = -0.036$, $SE = 0.011$, 95 % $CI = [-0.057, -0.014]$, $t = -3.19$, $p = 0.001$), indicating an asymmetry in mixing costs, with 102 ms for L1 and 55 ms for L2.

In the analysis of switch costs, the main effect of Trial type was significant ($b = 0.095$, $SE = 0.009$, 95 % $CI = [0.076, 0.113]$, $t = 10.53$, $p < 0.001$), showing that RTs were longer for switch trials (1399 ms) compared to non-switch trials (1273 ms), resulting in an averaged switch cost of 126 ms. The main effect of Language was not significant ($b = 0.003$, $SE = 0.015$, 95 % $CI = [-0.028, 0.034]$, $t = 0.18$, $p = 0.855$), suggesting no difference in naming speed between L1 (1331 ms) and L2 (1340 ms). Additionally, the interaction between Trial type and Language was significant ($b = 0.027$, $SE = 0.013$, 95 % $CI = [0.002, 0.051]$, $t = 2.12$, $p = 0.034$), indicating an asymmetry in switch costs, with 105 ms for L1 and 148 ms for L2.

To further examine the impact of bilingual typological differences on language control, we analyzed switch costs for the four types of classifier-noun phrases. No differences were found in L1 switch costs ($F(3, 69) = 0.184$, $p = 0.907$), L2 switch costs ($F(3, 66) = 1.625$, $p = 0.192$), or overall switch costs ($\chi^2(3) = 1.1$, $p = 0.777$).

3.3. The relationships between lexical control, phrasal control, and general cognitive control

(1) The relationships between lexical control and general cognitive control.

In analyzing the relationship between mixing costs and PBI, the interaction between Trial type and PBI was marginally significant ($b = 0.010$, $SE = 0.005$, 95 % $CI = [-0.000, 0.020]$, $t = 1.92$, $p = 0.055$), indicating that PBI tended to predict participants' lexical mixing costs, with higher PBI being associated with larger lexical mixing costs. Moreover, the interaction among Trial type, Language, and PBI was

Table 1

Mean reaction times (RT, ms, standard deviations) and ER (%) (standard deviations) in the lexical control task.

Index	Language	Single-language	Non-switch	Switch	Mixing cost	Switch cost
RT	L1	1017 (233)	1100 (244)	1215 (255)	83 (11)	115 (11)
	L2	1037 (236)	1082 (246)	1183 (277)	45 (10)	101 (31)
ER	L1	8 (0.28)	15 (0.25)	14 (0.35)	7 (-0.03)	-1 (0.1)
	L2	12 (0.32)	12 (0.33)	15 (0.35)	0 (0.01)	3 (0.02)

Table 2

Mean reaction times (RT, ms, standard deviations) and ER (%) (standard deviations) in the phrasal control task.

Index	Language	Single-language	Non-switch	Switch	Mixing cost	Switch cost
RT	L1	1176 (313)	1278 (334)	1383 (359)	102 (21)	105 (25)
	L2	1212 (324)	1267 (318)	1415 (393)	55 (-6)	148 (75)
ER	L1	15 (0.35)	18 (0.38)	16 (0.37)	3 (0.03)	-2 (-0.01)
	L2	15 (0.36)	14 (0.35)	16 (0.37)	-1 (-0.01)	2 (0.02)

significant ($b = 0.024$, $SE = 0.010$, 95 % $CI = [0.004, 0.045]$, $t = 2.31$, $p = 0.021$). This suggests that PBI may modulate the pattern of lexical mixing costs, with greater PBI being linked to larger differences in lexical mixing costs between L1 and L2.

In analyzing the relationship between switch costs and Stroop scores, the interaction between Trial type and Stroop score was not significant ($b = 0.002$, $SE = 0.006$, 95 % $CI = [-0.010, 0.013]$, $t = 0.32$, $p = 0.750$), indicating that the Stroop score did not predict lexical switch costs. Furthermore, the interaction among Trial type, Language, and Stroop score was not significant ($b = -0.011$, $SE = 0.012$, 95 % $CI = [-0.034, 0.012]$, $t = -0.91$, $p = 0.361$), suggesting that Stroop scores did not modulate the pattern of switch costs. The results using the new Stroop scores (calculated from $n-1$ congruent trials) are consistent with the above findings. Neither the interaction between Trial type and new Stroop score ($b = -0.006$, $SE = 0.006$, 95 % $CI = [-0.018, 0.006]$, $t = -1.02$, $p = 0.306$) nor the interaction among Trial type, Language, and new Stroop score ($b = -0.003$, $SE = 0.012$, 95 % $CI = [-0.026, 0.020]$, $t = -0.22$, $p = 0.826$) was significant.

(2) The relationships between phrasal control and general cognitive control.

In analyzing the relationship between mixing costs and PBI, the interaction between Trial type and PBI was significant ($b = -0.030$, $SE = 0.014$, 95 % $CI = [-0.060, -0.000]$, $t = -2.10$, $p = 0.047$), indicating that PBI tended to predict participants' lexical mixing costs, with higher PBI being associated with larger lexical mixing costs. Furthermore, the interaction among Trial type, Language, and PBI was not significant ($b = 0.005$, $SE = 0.011$, 95 % $CI = [-0.016, 0.027]$, $t = 0.47$, $p = 0.639$), suggesting that PBI did not modulate the pattern of phrasal mixing costs.

In analyzing the relationship between switch costs and Stroop scores, the interaction between Trial type and Stroop score was significant ($b = 0.017$, $SE = 0.006$, 95 % $CI = [0.004, 0.029]$, $t = 2.59$, $p = 0.010$), suggesting that Stroop scores predicted participants' phrasal switch costs, with higher Stroop scores being linked to larger switch costs. The interaction among Trial type, Language, and Stroop score was not significant ($b = 0.018$, $SE = 0.013$, 95 % $CI = [-0.007, 0.043]$, $t = 1.41$, $p = 0.158$), indicating that Stroop scores did not modulate the pattern of phrasal switch costs. Additionally, while the interaction between Trial type and new Stroop score was not significant ($b = -0.002$, $SE = 0.006$, 95 % $CI = [-0.014, 0.011]$, $t = -0.29$, $p = 0.775$), the interaction among Trial type, Language, and new Stroop score was significant ($b = 0.031$, $SE = 0.013$, 95 % $CI = [0.006, 0.056]$, $t = 2.44$, $p = 0.015$). This suggests that the new Stroop score predicts the asymmetry in switch costs (i.e., the efficiency of reactive language control), with higher Stroop scores being associated with larger differences in phrasal switch costs between L1 and L2.

4. Discussion

The current study set out to examine language control at different linguistic levels (lexical and phrasal) and their relationships with general cognitive control in a systematic fashion. Results and their implications were discussed in the following sections.

4.1. Lexical control and phrasal control: Mechanisms and interactions

As for the [Research Question 1](#), the lexical control results revealed symmetrical lexical switch costs, supporting the Inhibitory Control Model ([Green, 1998](#)). The ICM posits that L2 proficiency plays a crucial role in shaping bilinguals' language control mechanisms. Since bilinguals are typically more proficient in L1 than in L2, applying and releasing inhibition on L1 requires more cognitive resources than on L2, leading to higher switch costs for L1. As L2 proficiency increases, the difference in reactive language control demands between L1 and L2 diminishes. Hence, bilinguals apply less inhibition to the dominant L1, resulting in a more balanced allocation of reactive language control resources and symmetrical switch costs between the two languages.

Consistent with these findings, no difference in switch costs was observed between L1 and L2 among participants with high L2 proficiency ([Costa et al., 2006](#); [Costa & Santesteban, 2004](#); [Declerck et al., 2015](#); [Declerck & Philipp, 2015a](#); [Green & Abutalebi, 2013](#); [Meuter & Allport, 1999](#)).

Moreover, the larger L1 than L2 mixing costs suggest that bilinguals may engage proactive control mechanism more strongly when managing L1 than L2 ([Declerck, 2020](#); [Ma et al., 2016](#)). The language-specific selection hypothesis proposed by [Costa and Santesteban \(2004\)](#) provides an explanation for these findings. For proficient bilinguals, the selection thresholds for L1 and L2 lexicons are distinct. Although L1 lexical representations are generally more accessible, proficient bilinguals may consistently inhibit their dominant L1 to achieve a relatively balanced activation between the two languages before cross-linguistic competition. This process facilitates successful L2 generation and enhances overall language control performance. Hence, the asymmetry between L1 and L2 mixing costs may indicate differential engagement of proactive control mechanism across the two languages, potentially supporting more efficient language management in proficient bilinguals.

[Research Question 2](#) examined bilinguals' phrasal control. Results showed that both mixing costs and switch costs were asymmetrical. These findings regarding proactive phrasal control are consistent with those observed in lexical control. In contrast, the switch costs between L1 and L2 were also asymmetrical, with L2 switch costs higher than L1 switch costs. This suggests that the demand for reactive control should be greater for L2 than for L1.

However, the direction of the asymmetrical switch costs observed in this study diverges from that reported in previous research on language control (e.g., [Meuter & Allport, 1999](#); [Tarlowski et al., 2013](#); [Zhang & Wang, 2012](#)). In contrast, some studies on sentence production (e.g., [Sánchez et al., 2022](#); [Titus & Peeters, 2024](#)) have found a reversed asymmetry, with higher L2 switch costs. These studies employed paradigms with higher ecological validity, such as the voluntary switching paradigm ([Sánchez et al., 2022](#)) and the VR paradigm ([Titus & Peeters, 2024](#)). Since our study did not use these paradigms, we focus primarily on explaining the relative activation of L1 and L2 within the dual-language block. Given that L2 is relatively less dominant, it may be more challenging for bilinguals to control than L1. Consequently, bilinguals may adopt a mental strategy that enhances L2 activation during the dual-language block, balancing activation levels between L1 and L2, and potentially even reversing the dominant language from L1 to L2 ([Titus & Peeters, 2024](#)). As a result, controlling the "temporarily dominant" L2 may be more cognitively demanding than controlling L1, leading to higher L2 switch costs than L1.

The reversed asymmetry observed in our study may be influenced by the L1 background of the participants and the typological characteristics of Chinese classifiers. Compared to English, Chinese classifiers not only quantify objects but also categorize and individualize them ([Chen, 2021](#); [Her & Hsieh, 2010](#)). The uniqueness of Chinese classifiers, presents challenges for native English speakers in processing and learning Chinese classifiers ([Fu, 2007](#); [Hu, 2012](#); [Lin et al., 2022](#)). Therefore, bilinguals may pay more attention to the L2 Chinese phrase during the task to avoid errors. Analysis of the audio transcriptions revealed a higher proportion of naming errors and self-corrections, where participants often misnamed L1 phrases as L2 phrases (6.95 %). This suggests that bilinguals may maintain heightened activation levels for L2 phrases to ensure accurate selection and generation. A plausible explanation might be that applying and releasing inhibition for bilingual phrases requires more cognitive resources for L2 than for L1, leading to greater L2 phrasal switch costs.

However, further analysis revealed no differences among the four types of classifier-noun phrases, suggesting that language mapping complexity induced by the bilingual typological differences may have little influence on reactive phrasal control, thus consistent with the findings of [Zhang and Wang \(2012\)](#). This lack of effect might be

attributed to the relatively high L2 proficiency of the bilingual participants, who may no longer be sensitive to these typological distinctions. Hence, the observed reversed asymmetry may reflect the inherent complexity of the phrasal control task itself, in which L2 phrases are less dominant and less frequently used than L1 phrases, resulting in greater control demands for L2 phrasal control compared to L1 phrasal control.

Overall, to balance activation between the two languages, proficient bilinguals exert stronger proactive control mechanisms over their dominant L1 at both the lexical and phrasal levels. However, the shift from symmetry to asymmetry in phrasal control, as compared to reactive lexical control, may reflect the increased demands of linguistic complexity on reactive language control, potentially compromising the efficiency of bilinguals' reactive language control.

4.2. Language control at different linguistic levels: Dynamic engagement of general cognitive control

Research Question 3 focused on the relationships among lexical control, phrasal control, and general cognitive control. The results indicated that, for lexical control, bilinguals rely on general proactive cognitive control, as PBI predicted the lexical mixing costs. For phrasal control, the results suggested that bilinguals rely on the general reactive cognitive control. Specifically, traditional Stroop scores predicted the phrasal switch costs, while the new Stroop scores did not predict the phrasal switch cost but instead predicted the asymmetry of phrasal switch costs—an index reflecting the efficiency of reactive language control. Therefore, although different Stroop scores capture distinct aspects of reactive language control, the two findings together consistently support the engagement of domain-general reactive cognitive control in phrasal control. Additionally, PBI predicted phrasal mixing costs, indicating a reliance on general proactive cognitive control in phrasal control as well.

The Stroop scores did not predict lexical switch costs, suggesting that bilinguals might not rely on general reactive cognitive control in lexical control. This could be influenced by L2 proficiency. Prior studies involving less proficient bilinguals have found a significant correlation between lexical control and general reactive cognitive control (e.g., Jiao et al., 2019, 2021; Kang et al., 2017; Li et al., 2021; Liu et al., 2016; Timmer et al., 2019). Nevertheless, several studies involving bilinguals with high L2 proficiency failed to observe this relationship (e.g., Branzi et al., 2016; Calabria et al., 2012; Macizo et al., 2012; Prior & Gollan, 2013; Segal et al., 2019; Timmer et al., 2019; Wu et al., 2022). Hence, proficient bilinguals may develop a control mechanism distinct from that of less proficient bilinguals, and lexical control in highly proficient L2 bilinguals might not rely on general reactive cognitive control (Costa & Santesteban, 2004). In contrast, PBI predicted lexical mixing costs, suggesting the engagement of general proactive cognitive control in lexical control. In line with previous studies involving bilinguals with high L2 proficiency (e.g., Jylkkä et al., 2021; Liu et al., 2019; Prior & Gollan, 2013; Wang et al., 2022). These findings collectively indicated that increased L2 proficiency might shift lexical control from a reliance on general reactive cognitive control to general proactive cognitive control mechanisms.

In terms of the relationship between phrasal control and general cognitive control, we found that PBI predicted phrasal mixing costs, suggesting that highly proficient L2 bilinguals likely have developed a stable proactive control mechanism, enabling them to automatically adjust across linguistic levels. However, Stroop scores predicted phrasal switch costs, differing from the results observed in lexical control. This could result from the higher control demands needed for phrasal control.

Phrasal control involved the activation of both conceptual semantic nodes and specific phrase forms, which increases cognitive load of bilinguals (Zhang & Wang, 2012). A limited number of studies on phrasal control have shown that even highly proficient bilinguals exhibit asymmetrical switch costs (Chang et al., 2023; Tarlowski et al., 2013; Zhang & Wang, 2012). In phrasal control, bilinguals must allocate extra

cognitive resources to the reactive control of target phrases, increasing demands and affecting the engagement of general cognitive control (Chang et al., 2023; Jiang et al., 2023; Wang et al., 2022).

4.3. The hypothesis of a dynamic cognitive mechanism underlying language control

Drawing from our discussion of lexical control, phrasal control, and their relationships with general cognitive control, we proposed a hypothesis of a dynamic cognitive mechanism underlying language control, as illustrated in Fig. 2. The main idea conveyed by this model is that for a given L2 proficiency, bilinguals would dynamically select appropriate cognitive control mechanisms for language control according to the complexity of various linguistic levels. Moreover, increased language control demands prompts bilinguals to recruit general cognitive control across different linguistic levels, providing evidence for shared mechanisms between language control and general cognitive control (Jiang et al., 2023; Prior & Gollan, 2013; Wang et al., 2022).

For bilinguals with high L2 proficiency, the lexical representations of both L1 and L2 are well-established, leading to consistent lexical accessibility and selection speed (Costa et al., 2006; Costa & Santesteban, 2004; Wang et al., 2022). Consequently, bilinguals adjust the balance between proactive and reactive control mechanisms. They monitor lexical competition across both languages, applying stronger proactive control to the dominant L1 (Christoffels et al., 2007; Declerck, 2020).

During phrasal control, in contrast, bilinguals apply greater proactive language control over L1 phrases and apply more reactive language control over L2 (Chang et al., 2023; Tarlowski et al., 2013; Zhang & Wang, 2012). They recruit both general proactive and reactive control mechanisms to monitor and inhibition bilingual phrasal competition effectively, thereby optimizing the phrasal control performances (Declerck, 2020; Jiang et al., 2023; Prior & Gollan, 2013; Wang et al., 2022).

Hence, when L2 proficiency is kept relatively high, the key aspects of this model exhibit:

- (1) **Lexical Control:** Bilinguals would apply greater proactive language control to the L1 lexicon and recruit general proactive cognitive control.
- (2) **Phrasal Control:** Bilinguals would apply stronger proactive language control over L1 phrases and reactive language control over L2 phrases, simultaneously engaging both general proactive and reactive cognitive control.

4.4. Limitations and future research

This study is subject to certain limitations. Firstly, a key limitation is the relatively small sample size. Given the exploratory nature of this study, these findings should be considered preliminary and in need of further empirical validation. Future research with larger sample sizes will be necessary to confirm the present findings and to better assess the stability and generalizability of the observed effects.

Secondly, a variety of tasks are commonly employed to assess general cognitive control, each potentially capturing distinct aspects of this broader domain. The current study included only two such tasks. While both are representative and well-established in general cognitive control research, they may tap into only a narrow subset of the overall concept. Additionally, inconsistencies in scoring methods may lead to divergent findings. For example, when examining the relationship between phrasal control and general reactive cognitive control, two Stroop indices yielded discrepant patterns, despite both reflecting the same underlying principle. Future research should incorporate a broader array of tasks that target distinct dimensions of general cognitive control to provide a more comprehensive understanding. Moreover, stronger theoretical integration between general cognitive control frameworks and task-based indices is essential, along with greater attention to the

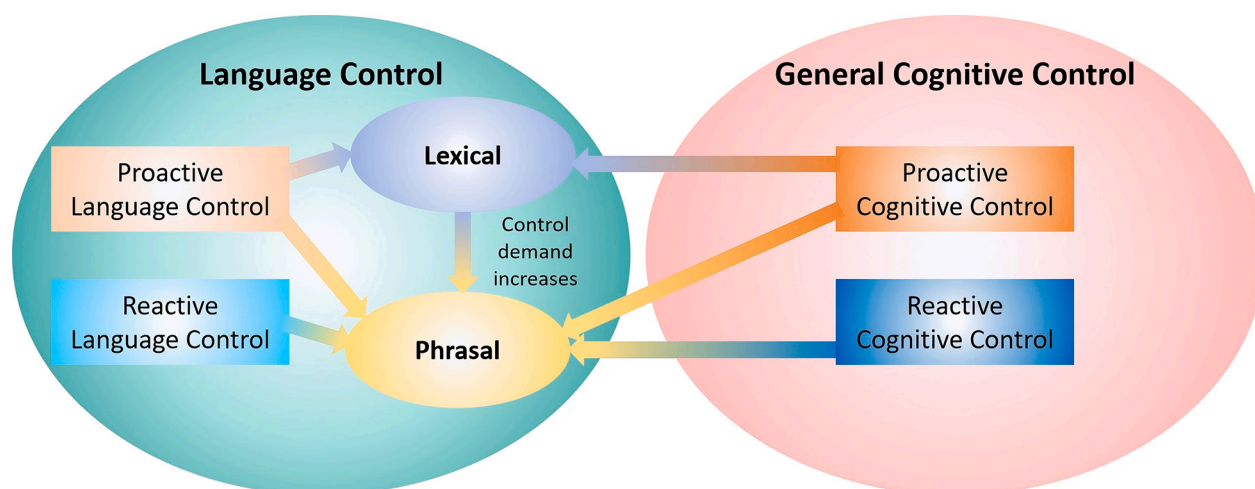


Fig. 2. The hypothesis of a dynamic cognitive mechanism underlying language control. Arrows indicate significant predictive relationships between the components. In lexical control, bilinguals recruit proactive language control along with general proactive cognitive control. In phrasal control, both proactive and reactive language control are engaged, as well as general proactive and reactive cognitive control.

reliability and construct validity of commonly used tasks and indices.

Thirdly, we examined language control at multiple linguistic levels, with participants' L2 proficiency controlled for and kept comparatively constant at an advanced level. As a result, the findings and hypotheses reflect the cognitive mechanisms underlying language control in advanced second language learners. Future research is encouraged to explore language control mechanisms across learners with various proficiency levels, investigate cognitive control differences among groups with varying proficiency, and to further examine how second language proficiency influences these mechanisms.

Finally, similar to those in this series of cued language switch paradigms, the language control indices employed in the present study might conceivably have been influenced by carryover effects. Specifically, the index of reactive language control, calculated by comparing switch and non-switch trials within the dual-language block, may reflect not only the immediate demands of language switching but also the lingering effects of control settings from the preceding trial. As such, the resulting switch cost may not purely represent reactive control, but instead may be confounded by residual control from the previous trial.

In contrast, the proactive control index, which is calculated by comparing non-switch trials in the dual-language block to those in the single-language block, is less susceptible to such carryover effects, as it does not involve trial-to-trial switching. This asymmetry underscores a key methodological challenge: disentangling proactive and reactive control processes, especially given the greater vulnerability of reactive control to residual activation from prior trials.

To address this issue and further refine our understanding of language control mechanisms, future research could consider to set longer inter-trial intervals (ITIs): Extending the time between trials may reduce carryover effects by allowing prior control settings to dissipate, leading to a purer measurement of reactive language control. Or future studies could employ the n-2 switching paradigms: These paradigms (e.g., comparing the second "A" in A-B-A versus C-B-A sequences, where A, B, and C represent three different languages) can help isolate the influence of current control demands from residual effects of previous settings (Declerck et al., 2015; Declerck & Philipp, 2018), providing a more accurate assessment of reactive control.

5. Conclusion

This study explored language control at the lexical and phrasal levels and their relationships with general cognitive control, proposing a hypothesis of a dynamic cognitive mechanism underlying language control

from the English-Chinese bilinguals. Our findings highlight three key notions. First, high-L2-proficiency bilinguals apply greater proactive control mechanisms to L1 at both linguistic levels, suggesting sustained inhibition of the dominant language L1. Second, reactive control demands vary between lexical and phrasal levels, influenced by L2 proficiency and linguistic complexity. Lastly, while general proactive cognitive control mechanism is recruited solely for lexical control, both general proactive and reactive cognitive control mechanisms are engaged for phrasal control. These results demonstrate how bilinguals adapt language control mechanisms at different linguistic levels and their relationships with general cognitive control under varying control demands. These were encompassed in the hypothesis of a dynamic cognitive mechanism underlying language control, offering latest insights into the interaction between control systems across language and cognitive domains, providing future research with tentative hypotheses for further verification.

CRedit authorship contribution statement

Qingwei Xue: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Qiping Wang:** Writing – review & editing, Methodology. **Junjie Wu:** Writing – review & editing, Methodology. **Luyao Chen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare none.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actpsy.2025.105277>.

Data availability

Data will be made available on request.

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